# Curvature Polarization Transport Gravity: The Hubble-Tension Bridge

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**Abstract**

The Hubble tension is usually described as a disagreement between an early-universe acoustic inference of the Hubble constant and a late-universe local distance-ladder measurement. Curvature Polarization Transport Gravity (CPTG) suggests a different interpretation. The two conventional values need not represent two incompatible physical expansion histories. Instead, they may be different comparison-layer projections of one native geometric branch. In this construction, the CPTG geometric branch lies between the acoustic/CMB comparison value and the local luminosity-distance comparison value. The acoustic projection maps the native branch downward, while the local luminosity projection maps it upward. The result is a two-sided bridge relation rather than a parameter-level contradiction.

## 1 Introduction

The conventional Hubble-tension statement compares a CMB-inferred value near

$$H_0^{\mathrm{Planck}} = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{1}$$

with a local distance-ladder value near

$$H_0^{\mathrm{SH0ES}} = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{2}$$

The Planck value is an acoustic-inference comparison quantity within the standard flat-ΛCDM interpretation of the CMB, while the local value is a luminosity-distance calibration quantity from the Cepheid/SN Ia distance ladder [1, 2].

Curvature Polarization Transport Gravity (CPTG) [3] does not begin by choosing one of these numbers as the single physical expansion branch. It separates the native geometric branch from the observational comparison layers. The preferred CPTG interpretation is therefore

$$\text{CMB acoustic-ruler projection} \neq \text{local luminosity-distance projection}. \tag{3}$$

The disagreement is treated as a projection split around one geometric source, not as two unrelated expansion histories.

The reason this split is natural in CPTG is that a geometric expansion curve is not a single observational object. CPTG separates three linked operations. Curvature sets the native geometric branch. Polarization determines how curvature stress is organized into an effective response. Transport determines how that response is carried into a particular observable channel. In compact form, an observational branch can be written schematically as

$$H_i(a) = \mathcal{P}_i(a)\mathcal{T}_i(a)H_{\mathrm{geom}}(a), \tag{4}$$

where $H_{\mathrm{geom}}(a)$ is the native CPTG geometric branch, $\mathcal{P}_i(a)$ is the polarization response for observational channel $i$, and $\mathcal{T}_i(a)$ is the corresponding transport response.

This does not mean CPTG contains arbitrary expansion histories. It means that different measurements can register different projected branches of the same geometric source. The acoustic branch is tied to early-universe sound-horizon transport and the CMB acoustic ruler. The local luminosity branch is tied to photon propagation, distance calibration, and local luminosity-distance transport. These branches can have different effective response speeds because curvature transport, polarization response, acoustic propagation, and luminosity propagation are not the same physical operation. The word "speed" is used here in this branch-response sense: it refers to how rapidly a given channel carries or records the curvature response, not to a new free particle velocity. This branch-response speed is not a local signal speed and is not introduced as a violation of relativistic causal propagation. It is an effective rate at which a given observational channel records or projects the curvature response.

For the Hubble-tension bridge, the relevant hierarchy is therefore

$$H_{\mathrm{ac}}(a) \longleftarrow H_{\mathrm{geom}}(a) \longrightarrow H_{\mathrm{loc}}(a), \tag{5}$$

where $H_{\mathrm{ac}}(a)$ is the acoustic/CMB comparison branch, $H_{\mathrm{geom}}(a)$ is the CPTG-native geometric branch, and $H_{\mathrm{loc}}(a)$ is the local luminosity-distance comparison branch. The article focuses on the present-epoch bridge between the three endpoints, while keeping the branch interpretation explicit.

The branch hierarchy can be summarized as follows.

| Layer | CPTG role | Observed form |
|---|---|---|
| Native curvature branch | Supplies the underlying geometric expansion source | $H_{\rm geom}(a)$ and $H_0^{(\pi)}$ |
| Polarization response | Organizes curvature stress into a channel response | $\mathcal{P}_i(a)$ |
| Transport response | Carries the polarized response into a measured channel | $\mathcal{T}_i(a)$ and $\Xi_i(a)$ |
| Acoustic/CMB projection | Records the branch through acoustic-ruler transport | $H_{0,\rm CMB}^{\rm CPTG} = H_0^{(\pi)}\sqrt{\mathcal{G}_T}$ |
| Local luminosity-distance projection | Records the branch through luminosity transport and calibration structure | $H_{0,\rm ladder}^{\rm CPTG} = H_0^{(\pi)}\mathcal{L}_{\rm loc}$ |

## 2 Parameter Set and Comparison-Layer Map

This section collects the numerical and structural quantities used in the Hubble-bridge construction. The purpose is not to introduce a multi-parameter fit. The native CPTG quantities are fixed by the geometric branch, while the observational endpoints are used as comparison anchors. The bridge is then expressed as a map between the native branch and the acoustic/CMB and local luminosity-distance comparison layers.

The numerical endpoints used here are fixed comparison anchors from Planck 2018 and Riess et al. 2022 [1, 2]. The purpose of the article is not to survey every current $H_0$ determination, but to test whether CPTG can organize the canonical CMB/local split through a two-sided comparison-layer map.

### 2.1 Native geometric quantities

The native CPTG branch is the geometric source branch of the present CPTG cosmology framework [3]. It is not identified directly with either the CMB-inferred comparison value or the local distance-ladder value. Its present-epoch normalization is

$$H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{6}$$

The branch exponent is

$$p_C = \pi. \tag{7}$$

The corresponding geometric weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \tag{8}$$

$$\mathcal{T}_\pi = \frac{1}{\pi}. \tag{9}$$

The native CPTG expansion branch is therefore

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \tag{10}$$

or equivalently,

$$H_{\text{geom}}(a) = H_0^{(\pi)}\sqrt{\left(1-\frac{1}{\pi}\right)+\frac{1}{\pi}a^{-\pi}}. \tag{11}$$

The fixed weight $\mathcal{T}_\pi$ should not be confused with the channel-dependent transport response $\mathcal{T}_i(a)$. The former defines the native $\pi$-branch weighting. The latter represents an observational comparison-layer response.

| Quantity | Value | Role |
|---|---|---|
| $p_C$ | $\pi$ | Native geometric branch exponent. |
| $\mathcal{B}_\pi$ | $1-1/\pi$ | Late/geometric branch weight. |
| $\mathcal{T}_\pi$ | $1/\pi$ | Native $\pi$-branch transport weight. |
| $H_0^{(\pi)}$ | 69.4162507897 | Native CPTG present-epoch geometric normalization in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. |

Table 1: Native CPTG branch quantities used in the Hubble-bridge construction.

### 2.2 Observational comparison anchors

The two external comparison anchors are the CMB/acoustic endpoint and the local distance-ladder endpoint [1, 2]. In this article they are not treated as two independent physical expansion histories. They are treated as observational comparison quantities.

$$H_0^{\text{Planck}} = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{12}$$

$$H_0^{\text{SH0ES}} = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{13}$$

The first is used as the acoustic/CMB comparison anchor. The second is used as the local luminosity-distance comparison anchor.

| Quantity | Value | Interpretation in this article |
|---|---|---|
| $H_0^{\text{Planck}}$ | $67.4 \pm 0.5$ | CMB/acoustic comparison endpoint. |
| $H_0^{\text{SH0ES}}$ | $73.04 \pm 1.04$ | Local luminosity-distance comparison endpoint. |

Table 2: External comparison anchors. Values are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

### 2.3 Projection factors

The acoustic/CMB projection maps the native CPTG branch downward by the fixed acoustic transport factor used in the CPTG comparison layer [3, 4]

$$\sqrt{\mathcal{G}_T} = 0.972074924351. \tag{14}$$

Thus,

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T}, \tag{15}$$

which gives

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{16}$$

The local luminosity-distance side is written as

$$H_{0,\mathrm{loc}}^{\mathrm{CPTG}} = H_0^{(\pi)}\mathcal{L}_{\mathrm{loc}}. \tag{17}$$

Using the SH0ES-like endpoint as the comparison anchor [2] gives

$$\mathcal{L}_{\mathrm{loc}} = \frac{73.04}{69.4162507897}, \tag{18}$$

so that

$$\mathcal{L}_{\mathrm{loc}} = 1.052203182527. \tag{19}$$

In this article, $\mathcal{L}_{\mathrm{loc}}$ is a required luminosity-transport factor and a derivation target. It is not claimed here as an independently derived validation parameter. The bridge succeeds as a theory only if this factor can be derived from CPTG luminosity transport, calibration geometry, environmental structure, or curvature-propagation response.

| Quantity | Value | Role |
| --- | --- | --- |
| $\sqrt{\mathcal{G}_T}$ | 0.972074924351 | Acoustic/CMB projection factor. |
| $H_{0,\mathrm{CMB}}^{\mathrm{CPTG}}$ | 67.4777967351 | CPTG acoustic/CMB comparison value. |
| $\mathcal{L}_{\mathrm{loc}}$ | 1.052203182527 | Required local luminosity-distance projection factor. |
| $H_{0,\mathrm{loc}}^{\mathrm{CPTG}}$ | 73.04 | CPTG local endpoint when anchored to the SH0ES-like comparison value. |

Table 3: Projection factors and projected endpoint values. $H_0$ values are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

### 2.4 Transport-response quantities

The branch structure also requires symbolic response functions [3]. These functions organize the theory language but are not numerically fitted in this article.

For an observational channel $i$, the projected branch may be written as

$$H_i(a) = \mathcal{P}_i(a)\mathcal{T}_i(a)H_{\rm geom}(a). \tag{20}$$

Here $\mathcal{P}_i(a)$ is the channel-dependent polarization response and $\mathcal{T}_i(a)$ is the channel-dependent transport response. These are structural response functions. They are not the same object as the fixed native branch weight $\mathcal{T}_\pi = 1/\pi$.

The geometric time coordinate is

$$dt_{\rm geom} = \frac{da}{aH_{\rm geom}(a)}. \tag{21}$$

A channel-observed comparison time is written as

$$dt_i = \Xi_i(a)dt_{\rm geom}, \tag{22}$$

where $\Xi_i(a)$ is the transport-time stretch factor for the observational branch. Therefore,

$$t_i(a) = \int_0^a \Xi_i(a')\frac{da'}{a'H_{\rm geom}(a')}. \tag{23}$$

The quantities $\mathcal{P}_i(a)$, $\mathcal{T}_i(a)$, and $\Xi_i(a)$ are the formal placeholders for branch-specific curvature response, transport speed, and observational time stretching. They encode the statement that acoustic propagation, luminosity transport, redshift mapping, and calibration structure need not project the same native geometric branch into identical comparison clocks.

| Quantity | Meaning | Status in this article |
|---|---|---|
| $\mathcal{P}_i(a)$ | Channel polarization response | Structural function, not fitted here. |
| $\mathcal{T}_i(a)$ | Channel transport response | Structural function, not the fixed weight $\mathcal{T}_\pi$. |
| $\Xi_i(a)$ | Transport-time stretch factor | Symbolic comparison-clock function. |
| $t_{\rm geom}(a)$ | Native geometric time | Defined by $H_{\rm geom}(a)$. |
| $t_i(a)$ | Channel-observed comparison time | Transport-projected time for channel $i$. |

Table 4: Transport-response quantities used to connect the branch theory to observational comparison layers.

### 2.5 Derived bridge diagnostics

The Hubble-bridge relation can now be summarized as

$$H_{0,\rm CMB}^{\rm CPTG} \longleftarrow H_0^{(\pi)} \longrightarrow H_{0,\rm loc}^{\rm CPTG}. \tag{24}$$

Numerically,

$$67.4778 \longleftarrow 69.4163 \longrightarrow 73.04. \tag{25}$$

The local endpoint is higher than the Planck comparison anchor by

$$\frac{73.04}{67.4} = 1.0837. \tag{26}$$

Relative to the native CPTG branch,

$$\frac{73.04}{69.4162507897} = 1.052203182527. \tag{27}$$

Equivalently, the native branch is lower than the local endpoint by

$$\frac{69.4162507897}{73.04} = 0.950385. \tag{28}$$

The distance-modulus shift associated with a Hubble-scale ratio is

$$\Delta\mu = 5\log_{10}\left(\frac{H_{\rm ref}}{H_{\rm loc}}\right). \tag{29}$$

Using the native CPTG branch as the reference gives

$$\Delta\mu_{\rm geom\to loc} = 5\log_{10}\left(\frac{69.4162507897}{73.04}\right) = -0.1105\,{\rm mag}. \tag{30}$$

Using the acoustic/CMB projected value as the reference gives

$$\Delta\mu_{\rm CMB\to loc} = 5\log_{10}\left(\frac{67.4777967351}{73.04}\right) = -0.1720\,{\rm mag}. \tag{31}$$

| Diagnostic | Value | Interpretation |
|---|---|---|
| $H_{0,\rm loc}/H_0^{\rm Planck}$ | 1.0837 | Local endpoint is about 8.4% above the Planck comparison anchor. |
| $H_{0,\rm loc}/H_0^{(\pi)}$ | 1.0522 | Local endpoint is about 5.2% above the native CPTG branch. |
| $H_0^{(\pi)}/H_{0,\rm loc}$ | 0.9504 | Native branch is about 5.0% lower in $H_0$ than the local endpoint; equivalently, the local endpoint corresponds to a shorter inverse-Hubble scale. |
| $\Delta\mu_{\rm geom\to loc}$ | $-0.1105$ mag | Luminosity-side shift relative to the native CPTG branch. |
| $\Delta\mu_{\rm CMB\to loc}$ | $-0.1720$ mag | Luminosity-side shift relative to the acoustic/CMB projection. |

Table 5: Derived bridge diagnostics. These values summarize the comparison-layer split around the native CPTG branch.

The parameter map therefore has two distinct roles. The fixed native quantities define the CPTG geometric branch. The comparison-layer quantities describe how acoustic and luminosity-distance observations project that branch into conventional $H_0$ summaries. The central claim of the article is not that the two observed endpoints are averaged, but that they occupy opposite projections of the same underlying curvature branch.

## 3 Native CPTG Geometric Branch

The native CPTG branch used in this article is the geometric $\pi$-branch introduced in the current CPTG cosmology program [3, 5]. The present-epoch native value is

$$H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{32}$$

The corresponding dimensionless expansion branch is

$$E_\pi(a)^2 = \left(1 - \frac{1}{\pi}\right) + \frac{1}{\pi}a^{-\pi}, \tag{33}$$

with

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \tag{34}$$

This branch is CPTG-native. It is not defined by forcing agreement with either the CMB endpoint or the local distance-ladder endpoint. Those endpoints are treated as comparison-layer projections of the branch.

## 4 Geometric Time and Transport-Observed Time

In CPTG, the native geometric branch defines the curvature clock of the theory [3]. This clock is not identical to every observational time inferred through a transport channel. Observations do not access the geometric branch directly. They access it through acoustic propagation, luminosity transport, redshift mapping, calibration structure, and polarization response. For this reason, CPTG distinguishes the native geometric time from transport-observed comparison time.

The native geometric time is defined by the CPTG geometric expansion branch,

$$dt_{\mathrm{geom}} = \frac{da}{aH_{\mathrm{geom}}(a)}. \tag{35}$$

Thus,

$$t_{\mathrm{geom}}(a) = \int_0^a \frac{da'}{a'H_{\mathrm{geom}}(a')}. \tag{36}$$

This is the time coordinate associated with the underlying curvature branch. It describes how the CPTG geometry accumulates expansion history before any observational comparison layer is applied.

A measurement channel does not necessarily sample this clock directly. If a signal propagates through a transport branch labeled by $i$, the comparison-layer time may be written as

$$dt_i = \Xi_i(a)dt_{\mathrm{geom}}, \tag{37}$$

where $\Xi_i(a)$ is the transport-time stretch factor for that observational channel. Equivalently,

$$t_i(a) = \int_0^a \Xi_i(a') \frac{da'}{a' H_{\rm geom}(a')}. \tag{38}$$

The factor $\Xi_i(a)$ is not introduced as an arbitrary clock correction. It represents the possibility that different branches of curvature response propagate, polarize, or calibrate through the geometry at different effective transport speeds. Acoustic rulers, luminosity distances, and local calibration chains therefore need not project the same geometric time into the same observational comparison time.

The important distinction is

$$t_{\rm geom}(a) \neq t_i(a) \tag{39}$$

in general, even though both are tied to the same underlying CPTG branch. This does not mean that physical clocks locally disagree. It means that an observationally inferred time or expansion scale is a transported comparison quantity. The inferred clock depends on the channel by which curvature information reaches the observer.

For the Hubble-tension bridge, this distinction is essential. The CMB/acoustic side samples the geometric branch through an acoustic-ruler transport projection. The local ladder side samples the same branch through a luminosity-distance and calibration projection. These two projections can stretch or compress the native geometric time differently,

$$t_{\rm ac}(a) = \int_0^a \Xi_{\rm ac}(a') \frac{da'}{a' H_{\rm geom}(a')}, \tag{40}$$

$$t_{\rm loc}(a) = \int_0^a \Xi_{\rm loc}(a') \frac{da'}{a' H_{\rm geom}(a')}. \tag{41}$$

Therefore, CPTG does not interpret the Hubble tension as evidence for two unrelated expansion histories. It interprets the tension as evidence that different observational channels transport the same geometric branch into different comparison-layer clocks and distance scales. In this view, time stretches with transport: the native curvature clock remains the geometric source, while the observed clock is the branch-specific projection of that source.

## 5 Acoustic/CMB Comparison Projection

The acoustic/CMB comparison layer maps the native branch downward through the CPTG transport factor [3]

$$\sqrt{\mathcal{G}_T} = 0.972074924351. \tag{42}$$

The CPTG acoustic-projected value is

$$
\begin{aligned}
H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} &= H_0^{(\pi)}\sqrt{\mathcal{G}_T} && (43)\\
&= 69.4162507897 \times 0.972074924351 && (44)\\
&= 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. && (45)
\end{aligned}
$$

This is close to the Planck comparison value [1],

$$H_0^{\mathrm{Planck}} = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{46}$$

The difference is

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} - H_0^{\mathrm{Planck}} = 0.0778\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{47}$$

which corresponds to

$$\frac{0.0778}{0.5} = 0.156\sigma \tag{48}$$

using the quoted Planck uncertainty.

The important point is interpretive. CPTG does not require the native branch itself to equal the CMB-inferred comparison value. The CMB value is the acoustic-ruler projection of the native branch.

## 6 Local Luminosity-Distance Projection

The local distance-ladder side is represented by a local luminosity/curvature-transport factor,

$$\mathcal{L}_{\mathrm{loc}}. \tag{49}$$

The comparison-layer relation is

$$H_{0,\mathrm{ladder}}^{\mathrm{CPTG}} = H_0^{(\pi)}\mathcal{L}_{\mathrm{loc}}. \tag{50}$$

For a SH0ES-like endpoint [2],

$$H_0^{\mathrm{SH0ES}} = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{51}$$

the required local luminosity-transport factor is

$$
\begin{aligned}
\mathcal{L}_{\mathrm{loc}} &= \frac{73.04}{69.4162507897} && (52)\\
&= 1.052203. && (53)
\end{aligned}
$$

Thus the ladder-side comparison value is

$$H_{0,\mathrm{ladder}}^{\mathrm{CPTG}} = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{54}$$

This should not be read as a claim that the distance ladder directly measures the full expansion curve at all redshifts. It is a local luminosity-distance projection anchored at the present endpoint.

## 7 The Hubble-Tension Bridge Relation

The CPTG bridge is the pair of relations

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T}, \tag{55}$$

$$H_{0,\mathrm{ladder}}^{\mathrm{CPTG}} = H_0^{(\pi)}\mathcal{L}_{\mathrm{loc}}. \tag{56}$$

Numerically,

$$67.4778 \longleftarrow 69.4163 \longrightarrow 73.04. \tag{57}$$

Equivalently,

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} \longleftarrow H_0^{(\pi)} \longrightarrow H_{0,\mathrm{ladder}}^{\mathrm{CPTG}}. \tag{58}$$

The usual Hubble tension is then rewritten as

$$\text{Hubble tension} \to \text{local luminosity-transport projection} \tag{59}$$

$$\neq \text{CMB acoustic-transport projection.} \tag{60}$$

CPTG converts the tension from a contradiction between two expansion histories into a two-sided projection relation around one native geometric branch.

## 8 Numerical Comparison

Using the fixed Planck and SH0ES comparison anchors [1, 2], the direct Planck-to-SH0ES ratio is

$$\frac{73.04}{67.4} = 1.0837, \tag{61}$$

so the local ladder endpoint is about 8.4% higher than the CMB-inferred endpoint. Conversely,

$$\frac{67.4}{73.04} = 0.9228, \tag{62}$$

so the CMB-inferred endpoint is about 7.7% lower than the local ladder endpoint.

Relative to the native CPTG branch, the acoustic/CMB projection is

$$\frac{H_{0,\mathrm{CMB}}^{\mathrm{CPTG}}}{H_0^{(\pi)}} = 0.9721, \tag{63}$$

while the local luminosity-distance projection is

$$\frac{H_{0,\mathrm{ladder}}^{\mathrm{CPTG}}}{H_0^{(\pi)}} = 1.0522. \tag{64}$$

| Quantity | Value | Role | Relative to $H_0^{(\pi)}$ |
|---|---|---|---|
| $H_0^{\mathrm{Planck}}$ | $67.4 \pm 0.5$ | CMB acoustic comparison value | 0.9710 |
| $H_{0,\mathrm{CMB}}^{\mathrm{CPTG}}$ | 67.4778 | CPTG acoustic projection | 0.9721 |
| $H_0^{(\pi)}$ | 69.4163 | Native CPTG geometric branch | 1.0000 |
| $H_{0,\mathrm{ladder}}^{\mathrm{CPTG}}$ | 73.04 | CPTG local luminosity projection | 1.0522 |
| $H_0^{\mathrm{SH0ES}}$ | 73.04±1.04 | Local distance-ladder comparison value | 1.0522 |

A side-by-side observational comparison is useful for separating direct comparison success from interpretive construction.

| Comparison | CPTG | Observed | Difference | Residual |
|---|---|---|---|---|
| Acoustic/CMB projection vs Planck | 67.4778 | $67.4 \pm 0.5$ | +0.0778 | $+0.156\sigma$ |
| Local luminosity projection vs SH0ES | 73.04 | 73.04±1.04 | 0.00 | by construction |

The local row is not an independent validation because $\mathcal{L}_{\mathrm{loc}}$ was chosen to represent the SH0ES-like endpoint. Its purpose is to state the required luminosity-side projection scale cleanly.

## 9 Distance-Modulus Target

Because local distance-ladder work is a luminosity-distance calibration problem, the ladder-side projection can also be expressed as a distance-modulus scale. Relative to the native CPTG branch,

$$\Delta\mu_{\mathrm{geom}\to\mathrm{lad}} = -5\log_{10}\left(\frac{73.04}{69.4162507897}\right) \tag{65}$$

$$= -0.1105\,\mathrm{mag}. \tag{66}$$

Relative to the acoustic/CMB-projected branch,

$$\Delta\mu_{\mathrm{CMB}\to\mathrm{lad}} = -5\log_{10}\left(\frac{73.04}{67.4777967351}\right) \tag{67}$$

$$= -0.1720\,\mathrm{mag}. \tag{68}$$

Thus the local luminosity projection corresponds to a target distance-modulus shift of roughly

$$\Delta\mu \simeq -0.11 \text{ to } -0.17\,\text{mag}, \tag{69}$$

depending on the comparison baseline.

This is a falsifiable scale. CPTG should not leave $\mathcal{L}_{\text{loc}}$ as a free endpoint multiplier. It must ultimately connect to luminosity transport, calibration geometry, curvature propagation, or another independently constrained CPTG mechanism.

## 10 Supporting Visualization

Figure 1 shows the three model-specific comparison curves used as a visualization of the bridge. The endpoint values and branch definitions follow the fixed comparison anchors and CPTG branch quantities used above [1, 2, 3]. The red curve is the acoustic/CMB comparison expansion curve, the orange curve is the native CPTG $\pi$-branch, and the green curve is the local distance-ladder comparison curve anchored to the local endpoint.

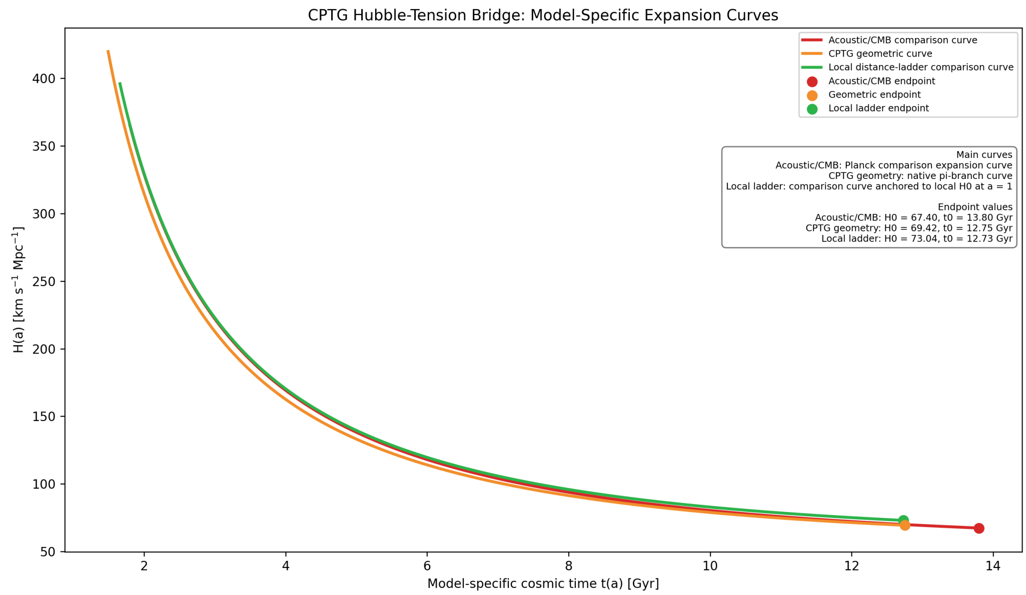


Figure 1: Model-specific expansion curves used as a visualization of the CPTG Hubble-tension bridge. The figure is not a fit and does not replace the bridge equations. It illustrates the separation between native CPTG geometry and observational comparison projections.

The plotted comparison curves are

$$H_{\text{ac}}(a) = 67.4\sqrt{0.315a^{-3} + 0.685}, \tag{70}$$

$$H_{\text{geom}}(a) = 69.4162507897\sqrt{\left(1 - \frac{1}{\pi}\right) + \frac{1}{\pi}a^{-\pi}}, \tag{71}$$

$$H_{\text{ladder}}(a) = 73.04\sqrt{0.315a^{-3} + 0.685}. \tag{72}$$

The local ladder curve is a comparison curve anchored to the local endpoint at $a = 1$. It is not a claim that Cepheid or SN Ia ladder methods directly measure the full redshift range.

## 11 Scope and Falsifiability

The present article is a bridge interpretation, not a final calibration proof. Its main claim is structural: the Hubble tension can be rewritten as a comparison-layer split around one native CPTG branch. The acoustic/CMB side is already tied to the locked CPTG comparison layer. The local luminosity side remains the open derivation target.

The key falsifiability condition is

$$\mathcal{L}_{\text{loc}} = 1.052203. \tag{73}$$

This number must not remain an arbitrary multiplier. A successful CPTG account must derive or constrain it from local luminosity transport, curvature propagation, environmental calibration structure, or a related CPTG mechanism. A successful luminosity-side derivation should predict not only the endpoint factor $\mathcal{L}_{\text{loc}}$, but also whether the corresponding distance-modulus shift depends on environment, wavelength, redshift range, host-galaxy structure, or calibration rung. If no such independent derivation is possible, the local side of the bridge remains descriptive rather than predictive.

The bridge also predicts the relevant calibration scale:

$$\Delta\mu \simeq -0.11 \text{ to } -0.17\,\text{mag}. \tag{74}$$

This gives a concrete target for future Cepheid/SN Ia calibration-layer tests.

## 12 Conclusion

CPTG does not remove the Hubble tension by averaging the CMB and local values, and it does not resolve the disagreement by declaring either measurement wrong. Instead, it places the conventional values on opposite sides of a native geometric branch. The CMB value is interpreted as an acoustic-ruler projection,

$$H_{0,\text{CMB}}^{\text{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T}, \tag{75}$$

while the local ladder value is interpreted as a luminosity-distance projection,

$$H_{0,\text{ladder}}^{\text{CPTG}} = H_0^{(\pi)}\mathcal{L}_{\text{loc}}. \tag{76}$$

The resulting bridge is

$$67.4778 \longleftarrow 69.4163 \longrightarrow 73.04. \tag{77}$$

In this view, the Hubble tension becomes evidence for a missing comparison-layer transport structure rather than evidence for two incompatible expansion histories. The next task is the derivation and observational testing of the local luminosity-transport factor.

## References


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[3] Carter L Glass Jr, Curvature Polarization Transport Gravity: A Unified Geometric Framework for Cosmic Structure and Expansion, DOI: 10.13140/RG.2.2.26030.68164.

[4] Carter L Glass Jr, CPTG Geometric Pi Branch: Comparison Coordinates, DOI: 10.13140/RG.2.2.15964.35201.

[5] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`.